

Name: Megan, Olivia, and Addie

Date: _____

Class: _____

MyDesign Portfolio

Element B Initial Template



The property owner is responsible for clearing that snow from the sidewalk.
The dog owner is responsible for clearing the dog

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Details

In NYC, property owners are legally responsible for clearing snow and ice from sidewalks in front of their buildings within a specific hour after snowfall stops. Failure to comply could result in fines from \$100-\$350, depending on the building size (medical excuses are allowed). Dept. of Sanitation clear streets, not sidewalks. Residents must report uncleared sidewalks for action to be taken.

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

Safety concerns because it delays increase slip and fall injuries.

Inconsistent enforcement because complaint-based systems can delay action.

Uneven results because some sidewalks are clear and some remain icy.

Prior Design Attempt #2 Municipal Sidewalk Plowing (Denmark Model)



Name: Megan, Olivia, and Addie**Date:** _____**Class:** _____sidercasino.dk

The Ultimate Bike Lane Snow Clearance ...

Details

The Copenhagen municipal government coordinates snow removal for sidewalks using small plows and pre-treatment salting. Preventative salting is applied before snow accumulation.

Include main details about this prior design attempt

Unlike NYC's private responsibility model, Copenhagen uses centralized city-managed crews.

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

- Higher municipal costs
- Difficult in very very dense cities like NYC
- Needs investment in specialized plowing equipment



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Prior Design Attempt #3 - <Snow Melting Machines (Japan Model)



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In regions of Japan like as Sapporo, cities use snow-melting machines that collect and melt snow using heated water systems. These systems reduce manual labor and maintain walkability even during heavy snowfall.

Analysis

- infrastructure - intensive
- Extremely high installation energy costs + time
- Can not change every street's strength and stiffness

It fails NYC design requirements of scalability across all 6 boroughs.

Prior Design Attempt #4 - <Heated Sidewalk Systems

Name: Megan, Olivia, and Addie**Date:** _____**Class:** _____ U.S.[U.S. Boiler Company](#)[Hydronic Snowmelt Systems 101 | US ...](#)

Details

Radiant snow-melt systems use electric or hydronic heating cables beneath pavement to melt snow automatically. Manufacturers such as Uponor promote these systems for commercial buildings and hospitals.

It prevents snow buildup entirely instead of removing it afterward.

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

- Requires significant energy consumption
- Realistic only for new developments or high-value areas



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Prior Design Attempt #5 -Small-Scale Sidewalk Snow Blowers

Ariens Classic 24-in Two-stage Self-propelled Gas Snow Blower Collection



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Commercial snow blowers are widely used by property owners and private contractors. Companies such as Ariens Company manufacture two-stage blowers capable of handling heavy snow.

→ These machines improve speed compared to manual shoveling and reduce labor strain.

Analysis

Provide evidence and explain what parts of this prior design attempt were unsuccessful at solving the defined problem and why based on design requirements.

- Still can rely on private individuals
- Does not guarantee timely removal
- Can not solve coordination issues
- Relies on private individuals